Maggie Makar

mmakar@mit.edu, mymakar.github.io

Education

Massachusetts Institute of Technology
Ph.D, Electrical Engineering and Computer Science, **Expected May 2021**

Massachusetts Institute of Technology
Master of Science, Electrical Engineering and Computer Science, **May 2017**
Thesis title: *Learning the probability of activation in the presence of latent spreaders*

University of Massachusetts, Amherst
B.S., Mathematics and Economics, *Summa Cum Laude*, **Feb 2013**

Employment

Research Assistant, Brigham and Women's Hospital	02/2013 - 08/2015
Research Assistant, Political Economy Research Institute	09/2012 - 12/2012
Research Intern, Donahue Institute,	08/2012 - 09/ 2012
Research Assistant, Center for Intelligent Information Retrieval	02/2012 -08/2012

Publications

Makar, M., Guttag, J., Wiens, J., "Learning the Probability of Activation in the Presence of Latent Spreaders". *To appear in AAAI-18*

Makar, M., Antonelli, J., Di, Q., Schwartz, J., Cutler, D., Dominici, F., "Estimating the Causal Effect of Fine Particulate Matter Levels on Death and Hospitalization: Are Levels Below the Safety Standards Harmful?". *Epidemiology. 2017 Sep 28(5): 627-634*

Obermeyer, Z., **Makar, M.**, Abujaber, S., Dominici, F., Block, S., Cutler, D., "Association between the Medicare hospice benefit and health care utilization and costs for patients with poor-prognosis cancer". *JAMA. 2014 Nov 12; 312(18): 1888-1896*

Makar, M., Ghassemi, M., Cutler, D., Obermeyer, Z., "Short-term mortality prediction for elderly patients using Medicare claims data". *International Journal of Machine Learning and Computing. 2015 Vol. 5(3): 192-197*

Powers, B., **Makar, M.**, Cutler, D., Obermeyer, Z., "Cost Savings Associated with Expanded Hospice Use Among Medicare Beneficiaries with Poor Prognosis Cancer". *J Palliat Med. 2015 May; 18(5):400-1*

Obermeyer, Z., Abujaber, S., **Makar, M.** , Stoll, S., Kayden, S., , Wallis, L., , Reynolds, T., "Emergency care delivery in 59 low- and middle-income countries: Systematic review". *WHO Bulletin. May 2015*

Obermeyer, Z., Powers, B., **Makar, M.**, Keating, N., Cutler, D., "Physician revealed preferences for hospice are strongly related to their patients' enrollment in hospice". *Health Affairs. 2015 Jun 1; 34(6):993-1000*

Obermeyer, Z., Clarke, A. , **Makar, M.**, Schuur, J., Cutler, D., "Emergency Care Use and the Medicare Hospice Benefit for Individuals with Cancer with a Poor Prognosis". *J Am Geriatr Soc. 2016 Feb; 64(2):323-9*

Workshops and poster sessions

A data-driven approach to predict daily risk of Clostridium difficile infection at two large academic healthcare centers. Infectious disease week. October 2017

Learning the probability of activation in the presence of latent spreaders. Statistics and Data Science conference at MIT. April 2017

Estimating the Causal Effect of Lowering Particulate Matter Levels Below the National Ambient Air Quality Standards on Health Outcomes. Joint Statistical Meeting (JSM). August 2016

Learning meaningful representations from medical claims data. Machine learning for healthcare workshops, NIPS. December 2015

Short-term mortality prediction for elderly patients using Medicare claims data. Max Planck Institute, July 2015

Invited talks

Inference on graphs with missing nodes. University of Michigan in Ann Arbor, July 2016

Association between the Medicare hospice benefit and health care utilization and costs for patients with poor-prognosis cancer. Harvard School of Public Health. October 2014

Teaching

Co-organizer, instructor: Whirlwind of Machine learning. Winter (IAP) 2017. MIT
Teaching assistant: Machine learning for Healthcare. Spring 2017. MIT

Awards

Ward McCarthy scholarship
School of Behavioral and Social Sciences scholarship
Economics Department Writing Award
Life Members Scholarship

Applicable Graduate Courses

- Machine learning
- Bayesian Modelling
- Computer Networks
- Cryptography and cryptanalysis

Computer and Language Skills

Computing: R, Python, Julia, LaTeX, and Stata
Languages: English, Arabic, and French